Quantitative Anomalies as Market Opportunities

A Statistical and AI-Assisted Dashboard for Equity and Forex Markets

Individual Report

Name: ______________________ Register No.: ________________

Submitted up to CIA 2

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1. Abstract

The Efficient Market Hypothesis assumes that prices absorb information instantly, that risk is broadly constant, and that returns cannot be forecast. Fifty years of empirical work says otherwise, and the documented departures — lagged macroeconomic transmission, volatility clustering, fat tails, and cointegration between related assets — are collectively called anomalies. This study treats each anomaly not as a defect in the theory but as a measurable opportunity, and builds a working system that finds them on live market data.

The project, **QuantAnomalies**, is a full-stack analytical dashboard with five statistical pillars: macro factor-and-lag regression, GARCH volatility modelling, Engle–Granger cointegration for forex pair trading, options pricing under four closed-form models, and the stochastic processes that underlie all of them. Thirteen deterministic detectors read these pillars and emit signals; a rule-based decision layer fuses the signals into a single directional verdict weighted by each detector's own statistical reliability; a locally hosted large language model then explains the verdict in plain English without ever being permitted to compute or alter a number.

Validated on the S&P 500 and a 45-pair forex universe over 2015–2025, the system finds that standardized macro factors explain 68.7% of monthly return variation (adjusted $R^2 = 0.585$), that GARCH(1,1) persistence reaches 0.994 with excess kurtosis of 15.8 and a Jarque–Bera p-value indistinguishable from zero, and that USDCHF and USDJPY are cointegrated at $p = 0.0075$ with a mean-reversion half-life of 69 days. The fused engine returns a cross-asset scan in 11 ms warm and a per-asset verdict in 4.5 ms. The principal contribution is architectural: statistics detect, deterministic rules decide, and the language model only explains — a separation that makes the output identical whether or not the model is running, and therefore auditable. The acknowledged limitation, stated deliberately, is that the results are in-sample: no walk-forward backtest, no multiple-testing correction on the cointegration sweep, and no transaction costs. These bound what the project currently claims and define the next block of work.

2. Introduction

Finance students meet the Efficient Market Hypothesis early and are told, quite correctly, that it is the right null hypothesis. Its three practical consequences — that prices reflect all available information, that volatility is a stable parameter, and that returns are unforecastable — are also the three assumptions that empirical finance has spent decades contradicting. Those contradictions are not scattered noise. They are repeatable, named, and published: Chen, Roll and Ross showed macroeconomic variables are priced factors; Mandelbrot, Engle and Bollerslev established that variance moves and clusters; Engle and Granger showed that individually unforecastable series can be tethered to each other.

The intellectual move this project makes is small but consequential. An anomaly is a statement that the model is wrong somewhere. Read from the other direction, it is a statement about where a careful observer might have an edge. A lagged macro effect is a positioning window. A volatility regime is a risk-sizing instruction. A stretched cointegrated spread is a market-neutral bet. The same statistic that a researcher reports as a rejection of a null is what a practitioner reports as a signal.

Three problems stand between that observation and anything usable. First, **fragmentation**: macro-factor work, volatility modelling and cointegration live in different sub-literatures and different toolchains, so almost nothing lets a user point at one arbitrary ticker and see all three at once. Second, **interpretation**: a persistence of 0.994 or a half-life of 69 days communicates nothing to a non-specialist, and the statistical literature is not written to fix that. Third, **aggregation**: even when several signals are available, combining them is usually done either by naive averaging, which manufactures false confidence out of disagreement, or by a machine-learning model whose output is a number nobody can interrogate.

This study addresses all three. It builds an integrated dashboard covering five statistical pillars on any user-supplied ticker; it adds a natural-language explanation layer that is constrained to explain rather than compute and is audited against the figures it was given; and it fuses the signals through an explicit, inspectable rule set in which conviction *falls* when detectors disagree and dissenting signals are displayed rather than averaged away.

The scope covered by this report runs from project inception to the CIA 2 presentation: the data pipeline and its validation, all five analysis pillars, the thirteen-detector decision layer, the fusion engine, the local AI layer, and the dashboard. Backtesting and the inferential hardening described in Section 7 are outside this scope and are stated as open work rather than claimed as done.

3. Literature Review

3.1 Efficient markets and their documented cracks

Fama (1970, 1991) sets the null: prices fully reflect available information, so risk-adjusted excess returns are unattainable. The anomalies literature is the record of where that null fails. Lo and MacKinlay (1988) statistically reject the random-walk model for U.S. weekly returns. Shiller (1981) shows that prices move far more than dividend fundamentals can justify — the excess-volatility finding. These results do not overturn the hypothesis so much as locate the places where structure survives, and those places are where this project looks.

3.2 Macro factors and lagged transmission

Arbitrage Pricing Theory (Ross, 1976) frames returns as driven by several systematic factors rather than one market beta. Chen, Roll and Ross (1986), in the canonical *Economic Forces and the Stock Market*, show that industrial production, inflation surprises, the term spread and default spreads are priced risk factors for equities. Fama and French (1989) demonstrate that dividend yield and term/default spreads forecast returns at business-cycle horizons — that is, macro effects operate **with a lag**, which is precisely the effect this project measures. To test direction rather than mere association, the study uses Granger causality (Granger, 1969), which asks whether a factor's past improves the forecast of returns beyond what returns' own past already provides. The macro pillar is therefore an OLS regression of returns on lagged macro factors, with Granger tests and AIC/BIC lag-depth comparison.

3.3 Volatility clustering and fat tails

Mandelbrot (1963), studying cotton prices, observed that large changes are followed by large changes and small by small, and that returns carry tails far heavier than the normal distribution admits. Engle (1982) formalised the first half as the ARCH model, letting conditional variance depend on recent squared shocks; Bollerslev (1986) generalised it to GARCH, and GARCH(1,1) is the specification fitted here. Cont (2001) catalogues the stylized facts — clustering, heavy tails, the leverage effect — that these models exist to capture. The properties are tested directly rather than assumed: Ljung–Box (1978) on squared returns for ARCH effects, and Jarque–Bera (1980) with Q–Q plots for non-normality. The practical reading is twofold — a high-volatility regime is a risk-management and option-pricing signal, and fat tails warn that variance-based risk measures understate danger.

3.4 Cointegration and pairs trading

Engle and Granger (1987) introduced cointegration: two non-stationary price series can share a stationary linear combination, so they remain tethered in the long run even while each is individually unpredictable. This is the statistical foundation of statistical arbitrage. Gatev, Goetzmann and Rouwenhorst (2006), the most-cited empirical treatment, show that a simple distance-based pairs rule earned significant excess returns on U.S. equities from 1962 to 2002, and document its decay as the strategy crowded. Vidyamurthy (2004) provides the standard practitioner pipeline — hedge ratio, spread, z-score, half-life — which is the pipeline implemented here.

3.5 Options pricing and the variance risk premium

Black and Scholes (1973) and Merton (1973) derived the closed-form European option price under geometric Brownian motion. Its most useful practical output is implied volatility: the volatility that equates model price to market price. The gap between option-implied volatility and subsequently realised or model-forecast volatility — the variance risk premium — is well documented and typically positive (Bakshi and Kapadia, 2003; Carr and Wu, 2009), meaning options are on average rich. Comparing market implied volatility with this project's own GARCH forecast is what bridges the volatility pillar to the options pillar and produces one of the thirteen detector signals.

3.6 Cross-sectional anomalies

Basu (1977) found that low price-to-earnings stocks earn higher risk-adjusted returns than high-P/E stocks — the value anomaly — later formalised with the size premium into the Fama–French three-factor model (1992, 1993) and extended to five factors (2015). Jegadeesh and Titman (1993) document momentum: 3–12 month winners continue to outperform over the following 3–12 months, confirmed across asset classes by Asness, Moskowitz and Pedersen (2013). Thaler (1987) surveys calendar effects. These support the momentum, relative-strength and seasonality detectors in the decision layer.

3.7 Statistical and AI tooling

Econometric work relies on peer-reviewed implementations in `statsmodels` (Seabold and Perktold, 2010) and the `arch` package for GARCH. The novel layer is the use of a **local** large language model to translate quantitative output into natural-language reasoning. The model is deliberately constrained to explain numbers it is given rather than compute them, mitigating the well-documented hallucination risk in language generation (Ji et al., 2023) and keeping the analyst note anchored to the deterministic statistics.

3.8 Data sources — assessment

A study is only as good as its instrument, and here the instrument is a set of financial-data APIs.

FRED, maintained by the Federal Reserve Bank of St. Louis, is an authoritative and widely cited repository of U.S. macroeconomic series with documented vintages and revision histories. Its limitation is intrinsic rather than technical: official statistics are released monthly with genuine publication lags and are subject to later revision, which is why this study models at monthly frequency and treats macro effects as lagged by construction.

Yahoo Finance, accessed through `yfinance`, provides daily OHLCV data across equities, indices, currencies, futures and crypto, with split- and dividend-adjusted prices and long history. The documented limitations are occasional data errors, silent changes to an unofficial API, possible survivorship bias for delisted names, and adjusted-price conventions that differ between vendors. This project encountered a more insidious variant, described in Section 5.2, which produced a caveat not found stated in the sources reviewed: with an unofficial client, the *access pattern* must be validated, not only the values.

DBnomics re-serves public economic datasets, including FRED, through a single open API, and is used here purely as a redundancy layer so that one provider outage cannot halt the study.

3.9 Research gap

The anomalies above are individually well studied, but three gaps motivate this work. **Integration** — few accessible tools bring macro-factor, volatility and cointegration analysis together for an arbitrary user-chosen asset. **Interpretation** — statistical output is opaque to non-specialists, and the remedy attempted here is a grounded explanation layer that is constrained to explain rather than compute and is audited against the figures supplied to it. **Actionability** — signals are fused into a single verdict weighted by each detector's own statistical reliability, with conviction falling on disagreement and dissenting signals displayed rather than averaged away, so the aggregation stays legible instead of becoming another opaque score.

4. Objectives of the Study

1. **Measure macro-driver opportunity.** Identify which macroeconomic forces — the volatility index, oil, gold, the dollar index, the 10-year yield, the Fed Funds rate, inflation and unemployment — move an asset's returns, and with what time delay, so that a lag can be read as a positioning window rather than a curiosity.
2. **Model the risk regime.** Estimate time-varying volatility via GARCH(1,1), test volatility clustering formally, and quantify the departure from normality, so that regime changes become an explicit risk-sizing input and tail risk is stated rather than assumed away.
3. **Detect mean-reversion opportunity.** Test a 45-pair forex universe for cointegration, construct the hedge ratio, spread and z-score for the strongest pair, generate entry and exit signals, and estimate the mean-reversion half-life.
4. **Price options and extract the variance risk premium.** Implement four closed-form pricing models on a shared engine with full Greeks, wire them to live market inputs, and compare market implied volatility against the project's own GARCH forecast.
5. **Synthesise the signals into one decision.** Fuse the detector outputs into a single directional verdict with explicit conviction, an independent risk axis, and a full audit trail — such that conviction falls on genuine conflict instead of averaging opposing evidence into a confident-looking middle.

6. **Explain the verdict in natural language, verifiably.** Use a locally hosted language model to narrate the decision under a strict constraint: it may not compute, rank, or invent a number, and every figure in its output is checked against the evidence it was supplied.
7. **Deliver accessibility and robustness.** Support any user-typed ticker and any forex combination on a fault-tolerant, cache-validated data pipeline, presented through an interactive dashboard.

5. Methodology

5.1 Research design

The design is quantitative, secondary-data, and computational. There is no questionnaire and no human sample; the instrument is a set of financial-data APIs and the sample is the live market history they return — daily price history from 2015 to 2025 across equities, indices, currencies, futures and crypto, together with monthly U.S. macroeconomic series. Analysis is conducted at daily frequency for volatility and pairs work and at monthly frequency for macro regression, matching the publication frequency of official statistics.

5.2 Data pipeline and validation

Prices are drawn from Yahoo Finance and macro series from FRED, with DBnomics as a fallback mirror, and everything is cached as CSV so the system runs offline once populated. Four validation practices are built in, each prompted by a fault actually encountered:

- **Range and sanity validation of every cached series.** This is what first exposed a corruption in which the volatility and oil caches held each other's data.
- **Serialized downloads.** Tracing that corruption revealed the mechanism: the `yfinance` client is not thread-safe, and concurrent `yf.download()` calls silently return one ticker's data labelled as another's. A deliberately threaded scan reproduced the fault on demand — three separate ticker pairs returned byte-identical frames. The failure is silent and plausible-looking, and no test that merely checks for missing data would catch it. Downloads now run behind a lock.
- **A cache staleness guard.** The download window's end date had been hardcoded, so once written, a cache was served indefinitely and every "latest price" in the application aged silently — at discovery, by 208 days. The end date now tracks the current date, a cached series older than seven days triggers a refetch, and if the refetch fails the stale cache is still served rather than raising, with the staleness exposed through a `data_freshness()` endpoint so it is visible rather than hidden. Monthly macro series are exempt, since their lag is genuine.
- **Fault-tolerant assembly.** If one factor source fails, that factor is skipped rather than breaking the entire study, and the omission is recorded.

5.3 Pillar 1 — Macro factor and lag regression

A monthly dataset is assembled containing the asset's return alongside eight standardized macro factors. OLS is estimated with time lags of zero to three months, so that both contemporaneous and delayed transmission are visible. Because the regressors are standardized, coefficients are comparable betas and the question "which driver matters most" has a fair answer. Granger-causality tests are run across the factor–lag grid, and a lagged-correlation heatmap supports visual inspection. Lag depth is compared by AIC and BIC.

5.4 Pillar 2 — GARCH and volatility clustering

A GARCH(1,1) model is fitted to daily log returns, producing a day-by-day conditional volatility series and a persistence estimate ($\alpha + \beta$). Clustering is tested with the Ljung–Box statistic on squared returns; normality is tested with Jarque–Bera and inspected with a Q–Q plot alongside skewness and excess kurtosis. The fitted model also produces the forward volatility forecast consumed by the options pillar.

5.5 Pillar 3 — Forex pair trading

Engle–Granger cointegration is tested across all pair combinations in the forex universe. For the strongest pair, a hedge ratio is estimated by regression on log prices, the spread is constructed and standardized to a z-score, and entry and exit signals are generated at thresholds of ± 2 and near zero respectively. The mean-reversion half-life is estimated from an AR(1) regression on spread changes.

5.6 Pillar 4 — Options pricing

Four closed-form models share a single engine and differ only in the carry term: Black–Scholes–Merton for equities, Garman–Kohlhagen for FX, Black-76 for futures, and Bachelier for normal dynamics. First- and second-order Greeks are computed analytically. A Cox–Ross–Rubinstein lattice handles American early exercise, and two smile-admitting models are implemented — the Merton jump-diffusion Poisson series and the Heston characteristic-function integral. Market wiring uses live spot, a continuously compounded rate derived from `^TNX`, the actual dividend yield, a GARCH volatility forecast aggregated over the option's life, and jump parameters fitted from the return distribution's own tails.

5.7 Pillar 5 — Stochastic processes

Simulation engines are implemented for the Wiener process, geometric Brownian motion, Ornstein–Uhlenbeck, Cox–Ingersoll–Ross, Merton jump-diffusion and Heston, each reporting theoretical moments beside empirical ones as a self-check. An Euler-versus-Milstein convergence study and a variance-reduction comparison (antithetic, control variate, and both) are included.

5.8 Coupling between pillars

The pillars are not independent exercises; four explicit couplings tie them into one system. The GARCH volatility forecast, aggregated over an option's life, *is* the sigma fed to Black–Scholes, on the reasoning that an option is a claim on future variance and a forecast is therefore the correct input rather than a historical average. Heston is seeded from GARCH by $\kappa = -252 \ln(\alpha + \beta)$, with θ the GARCH long-run variance and v_0 the current conditional variance, since both describe the same dynamics in discrete and continuous time. The pair spread is fitted as an Ornstein–Uhlenbeck process whose half-life $\ln(2)/\kappa$ independently reproduces the value the pairs module derives from AR(1). Finally, market implied volatility minus the GARCH forecast becomes the variance-risk-premium detector signal.

5.9 The decision layer

Three strictly separated layers constitute the design's principal claim.

Detection. Thirteen deterministic detectors emit signals with a severity in $[0, 1]$: trend, breakout, 12-1 momentum and relative performance on the price/trend axis; mean reversion (RSI plus displacement) and pairs opportunity on the mean-reversion axis; volatility regime, tail event and options mispricing on the volatility axis; macro dislocation; and volume anomaly, correlation regime and seasonality on the flow/context axis.

Decision. The signals are netted into one stance, conviction and position size. Each signal is weighted by reliability $\times$ severity, where reliability is derived from the detector's own statistics — cointegration p-value, macro R^2, GARCH sample size — and never invented. Redundant signals within a family are discounted by 0.45^{rank}, so three views of the same price move do not count as three confirmations. Direction (agreement $\times$ strength, with strength saturating through tanh) is separated from size, which is a risk overlay driven by risk-off signals and the volatility percentile. Conviction is demoted on genuine conflict rather than averaged, and a full audit trail names every step.

Fusion. A parallel engine answers where the evidence points and how firmly. Eight of the thirteen detectors feed it. Because detectors speak different direction vocabularies — "uptrend", "above_model", "long spread", "compressed", "rich" — a polarity map collapses them onto one bull/bear/neutral axis. Volatility and tail readings are routed to a separate risk axis so that a volatility spike can never masquerade as a directional call, and volatility mispricing is kept off both axes since expensive premium is a relative-value read. With weight $w = \text{severity} \times \text{reliability}$:

$$
\begin{aligned}
\text{tilt} &= \frac{\text{bull} - \text{bear}}{\text{bull} + \text{bear}} && \text{direction, } -1..+1 \\
\text{agreement} &= 1 - \frac{\min(\text{bull}, \text{bear})}{\max(\text{bull}, \text{bear})} && \text{how one-sided} \\
\text{mass} &= 1 - e^{-\text{total}/2} && \text{saturating evidence weight} \\
\text{conviction} &= \text{mass} \times (0.4 + 0.6 \times \text{agreement})
\end{aligned}
$$

Strength therefore beats count, nothing pins to 1.0 on noise, and a signal pointing the other way actively lowers conviction. This replaced an earlier linear form dominated by signal count that clamped to 1.0 as soon as six of anything appeared.

Explanation. The language model receives the detections and the computed decision as compact JSON and writes prose. It cannot compute, rank, invent a number, or change the stance, conviction or size; the system's output is identical whether or not a model is available. Three mechanisms make that verifiable rather than merely asserted: the prompt demands a `STANCE:` line before the prose, so the model's own labelled opinion lands within roughly ten tokens; that stance renders beside the computed stance, so divergence is visible rather than hidden; and an `unverified_numbers()` check tests every figure in the narrative against the supplied evidence, surfacing anything unmatched as a grounding badge in the interface.

5.10 Implementation and performance

The backend is FastAPI with pandas, numpy, statsmodels, arch and scipy, exposing 38 endpoints; the frontend is React 19 with Vite, Recharts and framer-motion. The language runtime is a local `llama.cpp` server running Qwen3-4B on an RTX 4050 via Vulkan, with a CPU fallback and a rules-only path if no model is running — nothing leaves the machine. Price detectors run broad and cheap, one cointegration sweep serves the whole FX basket, and GARCH is cached on (ticker, last data date) so it refits once per trading day regardless of request volume. A detector that fails is recorded in diagnostics rather than raised, so one dead factor costs one signal rather than the verdict.

5.11 Verification strategy

The numerical work is checked rather than trusted, with each check computed at request time and returned in the API response so it is visible in the interface. Put–call parity is a model-free arbitrage identity; Greeks are cross-checked against central finite differences; every Monte Carlo estimator is paired against the exact price for the same model; the binomial lattice is checked for $1/N$ error decay; and the Heston characteristic function is checked by driving vol-of-vol to zero, which must collapse the model to Black–Scholes.

6. Data Analysis and Results (to CIA 2)

All figures below were recomputed on 26 July 2026 on live data.

6.1 Pillar 1 — Macro factor regression (S&P 500, monthly)

Quantity	Value
R^2	0.687
Adjusted R^2	0.585
Granger tests run	32
Significant Granger relationships	6

Standardized macro factors explain roughly 69% of monthly return variation. Because the regressors are standardized, the coefficients are directly comparable, so the ranking of drivers is meaningful rather than an artefact of differing units. Six of thirty-two Granger tests reach significance, indicating that a subset of factors leads returns rather than merely co-moving with them — which is the lag that Objective 1 sought to measure. The gap between R^2 and adjusted R^2 reflects the cost of the lag structure in degrees of freedom and is reported rather than suppressed.

6.2 Pillar 2 — GARCH and the return distribution (S&P 500, daily)

Quantity	Value
Sample	2,905 trading days
GARCH(1,1) persistence ($\alpha + \beta$)	0.994
Excess kurtosis	15.8
Skewness	−0.65
Jarque–Bera p-value	≈ 0

Persistence of 0.994 means volatility shocks decay very slowly: a turbulent period predicts further turbulence for weeks, which is exactly the clustering that Mandelbrot described and that ARCH/GARCH was built to formalise. Excess kurtosis of 15.8 against a normal benchmark of 0, with negative skew, says that extreme moves — and particularly extreme *downward* moves — occur far more often than a Gaussian assumption admits. Jarque–Bera rejects normality decisively. The practical consequence is that variance-based risk measures on this series understate danger, and that a volatility regime is a legitimate risk-sizing input rather than a descriptive statistic.

One caveat is stated by the code itself and displayed in the interface: as persistence approaches 1, the GARCH long-run variance becomes ill-conditioned. Short-horizon forecasts from this fit are reliable; the long-run level is indicative only.

6.3 Pillar 3 — Cointegration and pairs (45-pair forex universe)

Quantity	Value
Strongest pair	USDCHF / USDJPY
Engle–Granger p-value	0.0075
Hedge ratio	−0.339
Mean-reversion half-life	69 days

The pair tested cointegrated on log prices, and the half-life of 69 days sets the horizon over which a stretched spread would be expected to close. The independent Ornstein–Uhlenbeck fit in Pillar 5 reproduces this half-life from $\ln(2)/\kappa$, which is a genuine cross-check: two different estimation routes, one AR(1)-based and one SDE-based, agree on the same reversion speed.

The result is reported with its qualification. The sweep searches many pair combinations and reports the best, so the p-value is not corrected for multiple comparisons; against a Bonferroni threshold over the 15 combinations in the reported subset, 0.0075 does not survive $0.05/15 = 0.00333$. What supports the finding is the economic rationale — both legs are USD-based safe-haven crosses and therefore share a common driver — rather than the p-value in isolation. Benjamini–Hochberg correction is the natural remedy and is scheduled work.

6.4 Pillar 4 — Options and the variance risk premium (AAPL)

Quantity	Value
Market implied volatility	33.1%
GARCH forecast volatility	28.8%
Variance risk premium	+4.3 points

The positive sign matches the documented literature (Bakshi and Kapadia; Carr and Wu): options are on average rich relative to subsequently realised volatility. Because the Heston model here is seeded from GARCH rather than calibrated to observed option prices, the comparison is not circular — calibrating to prices would have made the premium an artefact of the fit. The consequence is that the model smile is a prediction rather than a fit, which is the intended trade-off.

6.5 Numerical verification

Check	Method	Result
Put–call parity	model-free arbitrage identity	violation 0.0
Greeks	central finite differences, $O(h^2)$	max difference 4.6×10^{-7}
GBM Monte Carlo	vs analytic Black–Scholes	within 3 standard errors
Merton Monte Carlo	vs exact Poisson-weighted series	Δ 0.053, SE 0.039
Heston Monte Carlo	vs Fourier inversion of the CF	Δ 0.067, SE 0.050
Heston CF	vol-of-vol $\to 0$ collapses to Black–Scholes	Δ 3×10^{-6}
Binomial lattice	error against $1/N$	halves per step doubling
American call, no dividend	premium must be exactly zero	0.0
Longstaff–Schwartz	vs 800-step lattice	Δ 0.023, SE 0.029
Black–Scholes smile	implied-vol spread across strikes	0.000

The Heston check is the most informative. Driving vol-of-vol to zero must reduce the model to Black–Scholes, and agreement to six decimal places validates the complex arithmetic, the branch-cut handling and the quadrature simultaneously — against a formula derived by an entirely different route.

Variance-reduction efficiency was measured rather than asserted: plain Monte Carlo 1.00×, antithetic 1.97×, control variate 5.01×, and both combined **19.27**×.

6.6 Decision engine performance

Quantity	**Value**
Detectors implemented	13 (8 feeding the fusion engine)
Cross-asset feed, warm	11 ms
Per-asset verdict, warm	4.5 ms
Off-universe asset, cold	0.9 s
AI narration	$\approx$ 4 s, $\approx$ 28 tokens/sec, streaming, local

The warm figures are the result of the tiering described in Section 5.10 — a single cointegration sweep amortised across the FX basket and a GARCH cache keyed on the last data date, so the model refits once per trading day rather than once per request.

6.7 Data-pipeline findings

Two findings are worth reporting as results in their own right, because both are methodological rather than incidental. The first is the `yfinance` thread-safety fault described in Section 5.2: concurrent downloads silently mislabel one ticker's data as another's, reproduced on demand with three byte-identical frame pairs. The second is the frozen end date that had left every cached price series 208 days stale while the dashboard reported them as current. Both failures were silent and both produced plausible-looking output, which is precisely why they are worth stating: they were caught by range validation and a freshness check, not by any test for missing data. A third defect of the same character — a dividend-yield unit mismatch applying a 32% yield to a stock paying 0.32% — was found by the same route.

6.8 Interpretation against the objectives

Objectives 1 through 4 are met with quantified results: macro drivers are measured with their lags and a directional test, the risk regime is modelled and its non-normality quantified, a cointegrated pair is identified with its reversion horizon, and the variance risk premium is extracted. Objective 5 is met by the fusion engine, whose behaviour on conflicting evidence is the specific property that distinguishes it from averaging. Objective 6 is met and, more importantly, made checkable through the stance-beside-stance display and the number guardrail. Objective 7 is met: any ticker, any forex combination, on a pipeline that now validates rather than trusts its sources.

7. Limitations

These are stated deliberately, because they bound what the project currently claims.

1. **No backtest.** Signals are detected but never evaluated for profit and loss, Sharpe ratio or drawdown. Everything reported is in-sample. This is the largest gap: "opportunities" are not yet evidence.

2. **Multiple testing is uncorrected.** The cointegration sweep reports the best of many tests without a Benjamini–Hochberg or Bonferroni correction, as discussed in Section 6.3.
3. **OLS standard errors** are used on a monthly regression with autocorrelated residuals, where Newey–West HAC errors would be appropriate.
4. **Decision weights are stated judgement, not fitted** — deliberately, since fitting them on a few hundred monthly observations would overfit more convincingly than it would inform.
5. **No transaction costs** are modelled anywhere.
6. **GARCH long-run variance is ill-conditioned** at persistence near 1; short-horizon forecasts hold, the long-run level is indicative.
7. **A single 10-year rate serves all option maturities**, where a bootstrapped yield curve would be correct; FX policy rates are approximations.
8. **Heston is seeded from GARCH, not calibrated to option prices** — intentional, to keep the variance-risk-premium comparison non-circular, at the cost of the smile being a prediction rather than a fit.

8. Work Planned Beyond CIA 2

In priority order: a walk-forward backtest of the decision engine, which is the highest value next step and the one that converts detected signals into evidence; multiple-comparisons correction on pair selection; Newey–West HAC standard errors on the macro regression; transaction-cost modelling; a bootstrapped yield curve for option maturities; and KPSS and structural-break tests to complement ADF.

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